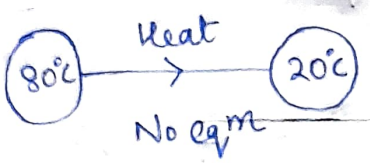
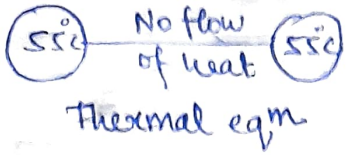
Thermodynamics

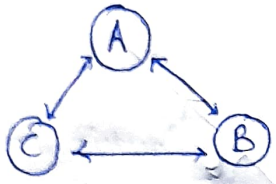
Thermal equilibrium: Two objects are in thermal eqm if they have equal temperature and there is no heat flow between them

Ex



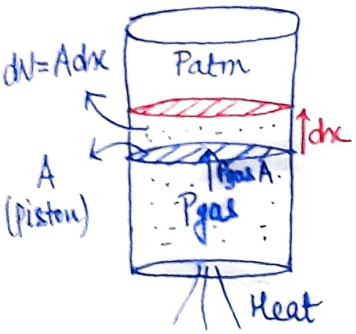


Zeroth law of Thermodynamics:



If A and B are in thermal eqm and A and C are also in thermal eqm, then B and C must also be in thermal eqm.

Work done by gas:



$dW_{gas} = \vec{F} \cdot \vec{ds}$

$= P_{gas} A dx \cos 0°$

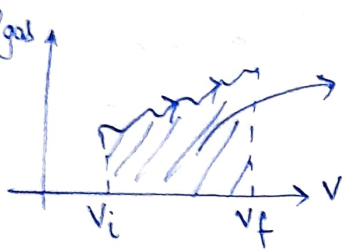
$= P_{gas} (Adx)$

$$\boxed{dW_{gas} = P_{gas}\, dV}$$

$$\boxed{W_{gas} = \int dW_{gas} = \int_{V_i}^{V_f} P_{gas}\, dV}$$

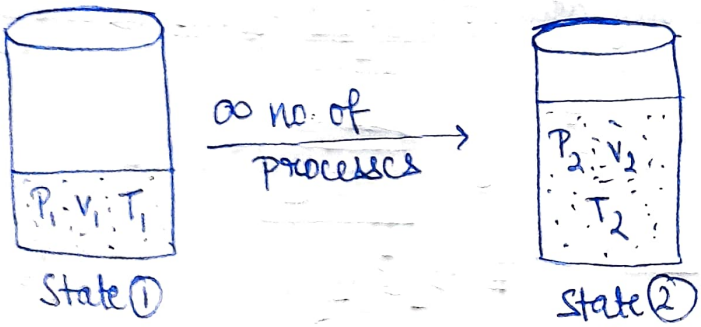
→ If V↑ (expand) ⇒ dV = +ve ⇒ W = +ve

→ If V↓ (compress) ⇒ dV = −ve ⇒ W = −ve



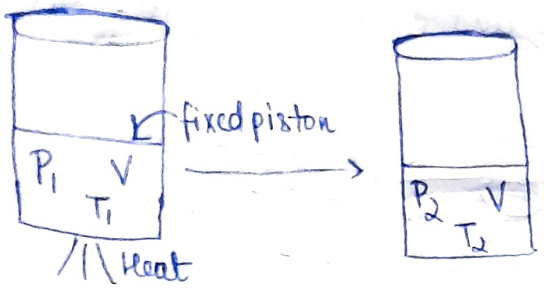
Area under Pgas/V gives Wgas.

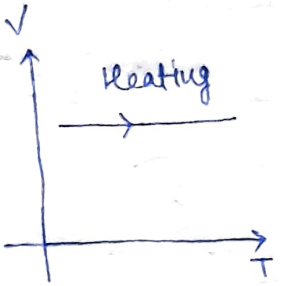
#

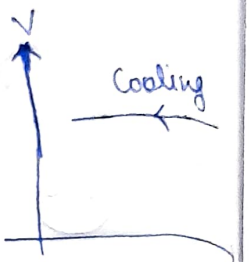


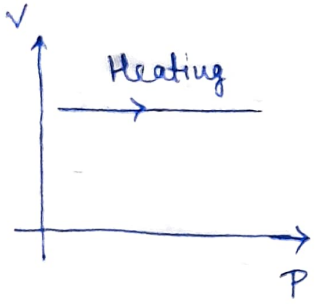
① Isochoric Process: $V = fix = const$
(isovolumetric)

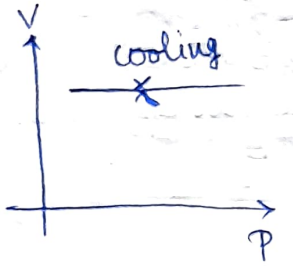
$\boxed{dV = 0}$

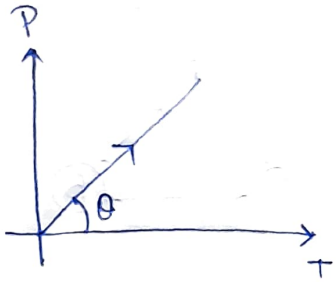












$PV = nRT \Rightarrow P = \left(\frac{nR}{V}\right)T \Rightarrow P \propto T$

$(y = mx)$

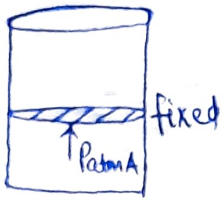
$\tan\theta = \frac{nR}{V}$

Work done in isochoric process:

$dW_{gas} = P_{gas} \cancel{dV}^{\,0}$

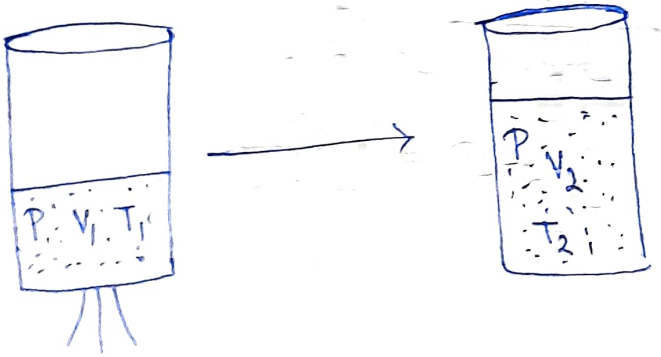
$dW_{gas} = 0$

$\therefore \boxed{W_{gas} = 0}$



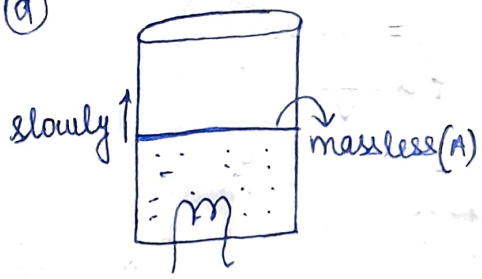
② Isobaric Process: $\boxed{P_{gas} = const \Rightarrow dP_{gas} = 0}$

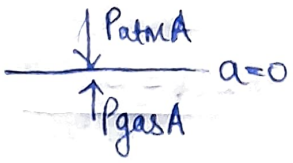
* Gas expands/compress, slowly against constant forces.



Ex

(a)



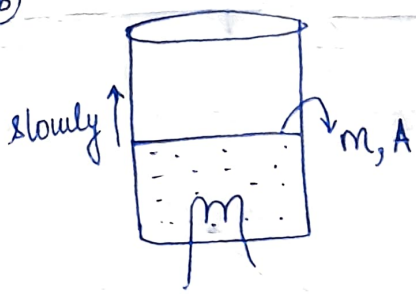


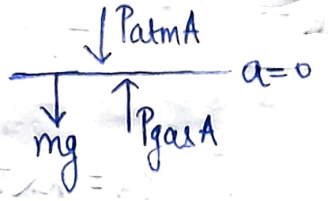
$P_{atm}\cancel{A} = P_{gas}\cancel{A}$

$\boxed{P_{gas} = P_{atm} = \text{const}}$

$\therefore$ isobaric process

(b)



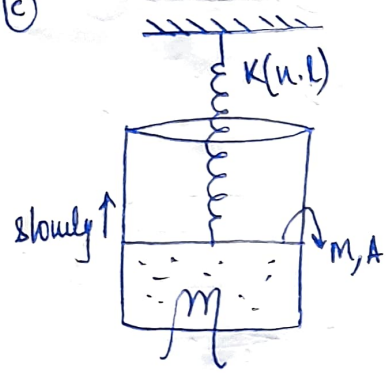


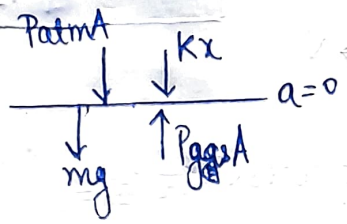
$P_{gas}A = P_{atm}\cancel{A} + mg$

$\boxed{P_{gas} = P_{atm} + \frac{mg}{A} = \text{const}}$

$\therefore$ isobaric process

(c)



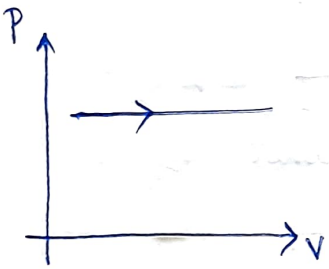


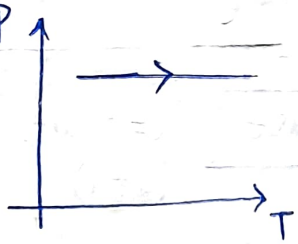
$P_{gas}A = P_{atm}A + mg + kx$

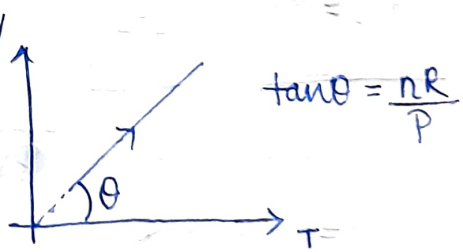
$P_{gas} = P_{atm} + \frac{mg}{A} + \frac{kx}{A} \neq \text{const}$

$\therefore$ Not isobaric process

Now, $PV = nRT \Rightarrow V = \left(\frac{nR}{P}\right)T \Rightarrow V \propto T$







Work done in isobaric process:

$$W_{gas} = \int_{V_i}^{V_f} P_{gas}\, dV$$

$$= P_{gas} \int_{V_i}^{V_f} dV$$

$$= P_{gas}(V)_{V_i}^{V_f} = P_{gas}(V_f - V_i)$$

$$\boxed{W_{gas} = P_{gas}\Delta V}$$

Also,

$$W_{gas} = P_{gas}V_f - P_{gas}V_i$$

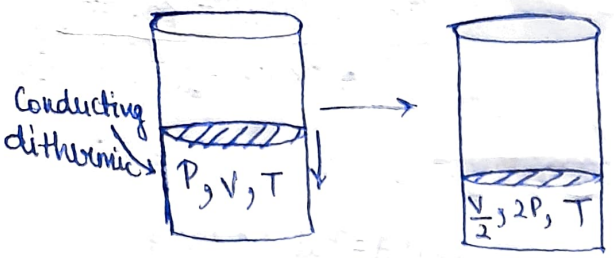
$$= nRT_f - nRT_i$$

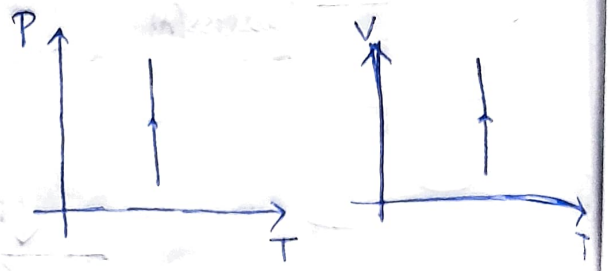
$$\boxed{W_{gas} = nR\Delta T}$$

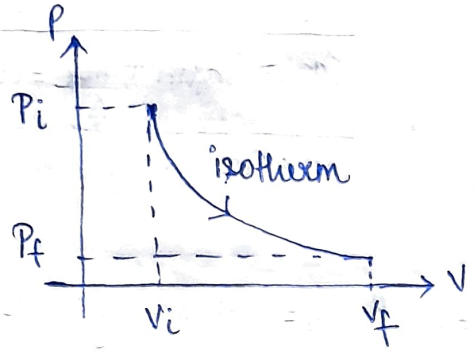
© Isothermal Process:

$T = \text{const}$
$dT = 0$

$PV = nRT$
$PV = \text{const}$
$P \propto \frac{1}{V}$







$PV = nRT$
$PV = \text{const}$

$\boxed{P_i V_i = P_f V_f}$

Work done in isothermal process:

$W_{gas} = \int P_{gas}\, dV$

$= \int \frac{nRT}{V}\, dV = nRT \int_{V_i}^{V_f} \frac{dV}{V} = nRT \ln(V)_{V_i}^{V_f}$

$\boxed{W_{gas} = nRT \ln\left(\frac{V_f}{V_i}\right)}$

$\boxed{W_{gas} = nRT \ln\left(\frac{P_i}{P_f}\right)}$

ⓓ Polytropic Process:

$\boxed{PV^a = \text{const}}$ a = polytropic const $(a \neq 1)$

$PV^a = \text{const} = K$

$\boxed{P_i V_i^a = P_f V_f^a = K}$

$W_{gas} = \int P\, dV$

$= \int K V^{-a}\, dV$

$= K \int_{V_i}^{V_f} V^{-a}\, dV$

$= K \left(\frac{V^{-a+1}}{-a+1}\right)_{V_i}^{V_f}$

$= K \frac{\left(V_f^{1-a} - V_i^{1-a}\right)}{1-a} = \frac{K V_f^{1-a} - K V_i^{1-a}}{1-a}$

$= \frac{P_f V_f^a V_f^{1-a} - P_i V_i^a V_i^{1-a}}{1-a}$

$\boxed{W_{gas} = \frac{P_f V_f - P_i V_i}{1-a}}$

$\boxed{W_{gas} = \frac{nR\Delta T}{1-a}}$

Change in internal energy of gas : It is a state function ie not dependant on process.

n moles
T

$$U = \frac{f}{2}nRT$$

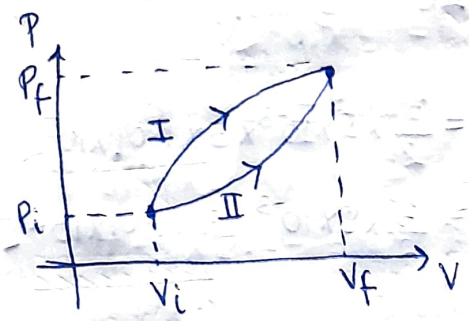
$$\boxed{\Delta U = \frac{f}{2}nR\Delta T}$$

$$\boxed{dU = \frac{f}{2}nR\,dT}$$

$$\Delta U = \frac{f}{2}nR(T_f - T_i) = \frac{f}{2}(nRT_f - nRT_i)$$

$$\boxed{\Delta U = \frac{f}{2}(P_f V_f - P_i V_i)}$$

Ex



$W_I > W_{II}$

$\Delta U_I = \Delta U_{II}$

First law of thermodynamics :

→ Basically energy conservation

Heat = Change in internal energy + Work done by gas.

$$\boxed{dQ = dU + dW}$$

① If heat is given ⇒ dQ = +ve
If heat is taken out ⇒ dQ = -ve

② If temp ↑ ⇒ dU = +ve
If temp ↓ ⇒ dU = -ve

③ If gas expands ⇒ dW = +ve
If gas compresses ⇒ dW = -ve

$$\therefore \boxed{\Delta Q = \Delta U + W}$$

① Isochoric Process :

W = 0

$$\Delta U = \frac{f}{2}nR\Delta T = \frac{f}{2}nR(T_2 - T_1)$$

$$\Delta U = \frac{f}{2}(nRT_2 - nRT_1)$$

$$\Delta U = \frac{f}{2}(P_f V_f - P_i V_i)$$

$$\Delta U = \frac{f}{2}V(P_f - P_i) = \frac{f}{2}V\Delta P$$

$$\Delta Q = \Delta U + W$$

$$= \frac{f}{2}V\Delta P + 0$$

$$\boxed{\Delta Q = \frac{f}{2}V\Delta P}$$

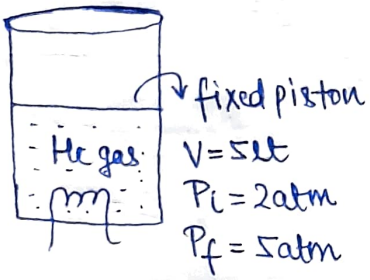
Q 48 gm O_2 gas is heated in a rigid container such that its temperature changes from 27°C to 37°C. Find ① W.d by gas ② ΔU ③ Heat given to gas

① Isochoric process ⇒ W = 0

② $\Delta U = \frac{f}{2} nR\Delta T = \frac{5}{2} \times \frac{48}{32} \times \frac{25}{3} \times 10 = \frac{625}{2} J = 312.5\ J$

③ $\Delta Q = \Delta U + W = 312.5 + 0 = 312.5\ J$

Q



Find ① W ② ΔU ③ ΔQ

① W = 0

② $\Delta U = \frac{f}{2} V\Delta P = \frac{3}{2} \times 5 \times 10^{-3} \times 3 \times 1.01 \times 10^5\ J$

③ $\Delta Q = \Delta U + \cancel{W}^{0} = \frac{3}{2} \times 5 \times 10^{-3} \times 3 \times 1.01 \times 10^5\ J$

② Isobaric Process:

$W = nR\Delta T = P\Delta V$

$\Delta U = \frac{f}{2} nR\Delta T = \frac{f}{2}(PV_f - PV_i)$

$\Delta U = \frac{f}{2} P(V_f - V_i) = \frac{f}{2} P\Delta V$

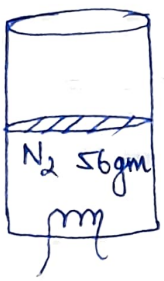
$\Delta Q = \Delta U + W$

$= \frac{f}{2} nR\Delta T + nR\Delta T$

$\boxed{\Delta Q = \left(\frac{f}{2} + 1\right) nR\Delta T}$

$\boxed{\Delta Q = \left(\frac{f}{2} + 1\right) P\Delta V}$

Q If gas is expanding slowly isobarically so that temperature of N_2 gas changes from 25°C to 35°C. Find ① W ② ΔU ③ ΔQ

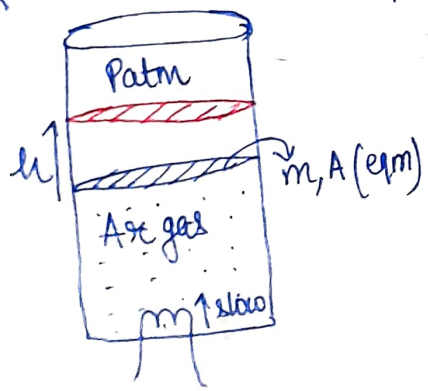


① $W = nR\Delta T = \frac{56\ gm}{28\ gm/mol} \times \frac{25}{3} \frac{J}{K\text{-}mol} \times 10\ K = \frac{500}{3} J$

② $\Delta U = \frac{f}{2} nR\Delta T = \frac{5}{2} \times \frac{500}{3} = \frac{2500}{6} J = \frac{1250}{3} J$

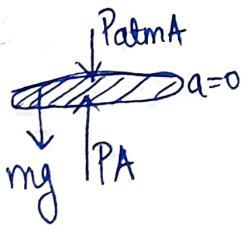
③ $\Delta Q = \Delta U + W = \frac{1250}{3} + \frac{500}{3} J = \frac{1750}{3} J$

Q



Piston moves slowly by distance h.

Find ① W ② ΔU ③ ΔQ



$PA = PatmA + mg$

$\boxed{P = Patm + \frac{mg}{A}}$

① $W = P\Delta V = \left(P_{atm} + \frac{mg}{A}\right)Ah = P_{atm}Ah + mgh$

or

using WET,

$W_g + W_{atm} + W_{gas} = \Delta KE$

$-mgh - P_{atm}Ah + W_{gas} = 0$

$W_{gas} = P_{atm}Ah + mgh$

② $\Delta U = \frac{f}{2} P\Delta V = \frac{3}{2}\left(P_{atm} + \frac{mg}{A}\right)Ah$

③ $\Delta Q = \Delta U + W = \left(P_{atm} + \frac{mg}{A}\right)Ah \left\{\frac{3}{2} + 1\right\}$

$\Delta Q = \frac{5}{2}\left(P_{atm} + \frac{mg}{A}\right)Ah$

③ Isothermal Process:

$W = nRT \ln \frac{V_f}{V_i} = nRT \ln \frac{P_i}{P_f}$

$\Delta U = 0 \quad (\Delta T = 0)$

$\Delta Q = \Delta U + W = nRT \ln\left(\frac{V_f}{V_i}\right)$

∴ isothermal heating ⇒ expansion

isothermal cooling ⇒ compression

Q 88gm of CO_2 gas is heated at const. temperature $T = 27°C$, such that its volume becomes thrice. Find ① W ② ΔU ③ ΔQ.

① $W = nRT \ln\left(\frac{V_f}{V_i}\right) = \frac{88}{44} \times \frac{25}{3} \times 300 \ln\left(\frac{3V}{V}\right) = 5000 \ln 3 \text{ J}$

② $\Delta U = 0$

③ $\Delta Q = \Delta U + W = 5000 \ln 3 \text{ J}$

④ Polytropic Process: $PV^a = \text{const}$

$W = \frac{nR\Delta T}{1-a} = \frac{P_f V_f - P_i V_i}{1-a}$

$\Delta U = \frac{f}{2} nR\Delta T = \frac{f}{2}(P_f V_f - P_i V_i)$

$\Delta Q = \Delta U + W = nR\Delta T\left(\frac{f}{2} + \frac{1}{1-a}\right) = (P_f V_f - P_i V_i)\left\{\frac{f}{2} + \frac{1}{1-a}\right\}$

Q 8gm He gas is heated to increase the temp by 10°C through process $P^2V = \text{const}$

Find ① W ② ΔU ③ ΔQ

$P^2V = \text{const}$

Taking sq. root both sides

$PV^{1/2} = \text{const} \quad \{PV^a = \text{const}\}$

$\Rightarrow a = 1/2$

① $W = \frac{nR\Delta T}{1-a} = \frac{\frac{8}{4} \times \frac{25}{3} \times 10}{1 - 1/2} = \frac{1000}{3} \text{ J}$

② $\Delta U = \frac{f}{2} nR\Delta T = \frac{3}{2} \times \frac{8}{4} \times \frac{25}{3} \times 10$

$= 250 \text{ J}$

③ $\Delta Q = \Delta U + W = 250 + \frac{1000}{3} \text{ J}$

$\Delta Q = \frac{1750}{3} \text{ J}$

Q O_2 gas is expanded through process $TV = const$ such that volume changes from $V = 2m^3$ to $4V = 8m^3$. Initial pressure of gas $= 2 \times 10^5 N/m^2$
Find ① W ② ΔU & ③ ΔQ

$TV = const$

We know, $PV = nRT$

$\Rightarrow T = \frac{PV}{nR}$

So, $\frac{PV}{nR} \cdot V = const$

$\Rightarrow PV^2 = const$

$\therefore \boxed{a = 2}$

Now, as $PV^2 = const$

$P_i V_i^2 = P_f V_f^2$

$2 \times 10^5 \times 2^2 = P_f \times 8^2$

$P_f = \frac{2 \times 10^5 \times 4}{64 \times 8}$

$P_f = \frac{10^5}{8} N/m^2$

① $W = \frac{P_f V_f - P_i V_i}{1-a}$

$= \frac{\frac{10^5}{8} \times 8 - 2 \times 10^5 \times 2}{1-2}$

$= 3 \times 10^5 J$

② $\Delta U = \frac{f}{2}(P_f V_f - P_i V_i)$

$= \frac{5}{2}\left(\frac{10^5}{8} \times 8 - 2 \times 10^5 \times 2\right) = -\frac{5}{2} \times 3 \times 10^5$

$\Delta U = -7.5 \times 10^5 J$

③ $\Delta Q = \Delta U + W = -7.5 \times 10^5 + 3 \times 10^5 = -4.5 \times 10^5 J$

Molar heat capacity (c):



$\boxed{C = \frac{\Delta Q}{n \Delta T} = \text{molar heat capacity}}$

→ It is dependent on process
→ Unit = Joule/mol-kelvin

① Isochoric Process ($V = fix$):

$\Delta Q = \Delta U + \cancel{W}^{0}$

$\Delta Q = \frac{f}{2} nR\Delta T$

$C_V = \frac{\Delta Q}{n\Delta T} = \frac{\frac{f}{2} \cancel{n} R \cancel{\Delta T}}{\cancel{n} \cancel{\Delta T}} = \frac{f}{2} R$

↳ Molar heat capacity for constant volume.

∴ For monoatomic ⇒ $C_V = 3/2 R$
For diatomic/Poly (linear) ⇒ $C_V = \frac{5}{2} R$
For Poly (non-linear) ⇒ $C_V = 3R$

② Isobaric Process ($P = fix$):

$\Delta Q = \Delta U + W = \frac{f}{2} nR\Delta T + nR\Delta T$

$\Delta Q = \left(\frac{f}{2} + 1\right) nR\Delta T$

$C_P = \frac{\Delta Q}{n\Delta T} = \frac{\left(\frac{f}{2} + 1\right) \cancel{n} R \cancel{\Delta T}}{\cancel{n} \cancel{\Delta T}} = \left(\frac{f}{2} + 1\right) R$

↳ Molar heat capacity for constant pressure

∴ For monoatomic $= \frac{5}{2} R$
For diatomic/Poly (linear) $= \frac{7}{2} R$
For Poly (non-linear) $= 4R$

Meyer's Relation:

$$C_p - C_v = \left(\frac{f}{2}+1\right)R - \frac{f}{2}R$$

$$\boxed{C_p - C_v = R}$$

Ratio of specific heats:

$$\gamma = \frac{C_p}{C_v} = \frac{\left(\frac{f}{2}+1\right)R}{\frac{f}{2}R} = \frac{f+2}{f}$$

$$\boxed{\gamma = \frac{C_p}{C_v} = 1 + \frac{2}{f}}$$

Type of molecule	Example	f	C_v	C_p	$\gamma = \frac{C_p}{C_v} = 1+\frac{2}{f}$
Monoatomic	He, Ne, Ar	3	$\frac{3}{2}R$	$\frac{5}{2}R$	$\gamma = \frac{5}{3} = 1.67$
Diatomic / Poly (linear)	O_2, N_2, Cl_2	5	$\frac{5}{2}R$	$\frac{7}{2}R$	$\gamma = \frac{7}{5} = 1.4$
Poly (Non-linear)	NH_3, H_2O	6	$3R$	$4R$	$\gamma = \frac{4}{3} = 1.33$

$$\boxed{C_{v_{mix}} = \frac{n_1 C_{v_1} + n_2 C_{v_2}}{n_1 + n_2}} \qquad \boxed{C_{p_{mix}} = \frac{n_1 C_{p_1} + n_2 C_{p_2}}{n_1 + n_2}} \qquad \boxed{\gamma_{mix} = \frac{C_{p_{mix}}}{C_{v_{mix}}} = \frac{n_1 C_{p_1} + n_2 C_{p_2}}{n_1 C_{v_1} + n_2 C_{v_2}}}$$

Q Mixture: 8gm He + 32gm O_2. Find ① $C_{v_{mix}}$ ② $C_{p_{mix}}$ ③ γ_{mix}

Moles of He = $8/4 = 2$ mol

Moles of O_2 = $32/32 = 1$ mol

① $C_{v_{mix}} = \frac{2 \times \frac{3}{2}R + 1 \times \frac{5}{2}R}{2+1} = \frac{11R}{6}$

② $C_{p_{mix}} = \frac{2 \times \frac{5}{2} + 1 \times \frac{7}{2}R}{2+1} = \frac{17R}{6}$

③ $\gamma_{mix} = \frac{C_{p_{mix}}}{C_{v_{mix}}} = \frac{17}{11}$

Note:

$$\boxed{\gamma_{mix} = 1 + \frac{2}{f_{mix}}} \qquad \boxed{f_{mix} = \frac{n_1 f_1 + n_2 f_2}{n_1 + n_2}}$$

Also,

$C_p - C_v = R$ — ①

$\gamma = \frac{C_p}{C_v}$ — ②

$C_p = \gamma C_v$

Put in ①

$\gamma C_v - C_v = R$

$$\boxed{C_v = \frac{R}{\gamma - 1}}$$

Similarly,

$$\boxed{C_p = \frac{\gamma R}{\gamma - 1}}$$

Alternate form of change in internal energy:

$$\Delta U = \frac{f}{2} n R \Delta T$$

$$\boxed{\Delta U = n C_v \Delta T} \quad \left\{C_v = \frac{f}{2}R\right\}$$

$$\boxed{\Delta U = \frac{nR\Delta T}{\gamma - 1}} \quad \left\{C_v = \frac{R}{\gamma - 1}\right\}$$

③ Isothermal process ($T=fix \Rightarrow \Delta T=0$):

$$C = \frac{\Delta Q}{n\Delta T} = \text{not defined}$$

④ Polytropic Process ($PV^a = const$):

$$W = \frac{nR\Delta T}{1-a} \qquad \Delta U = \frac{f}{2}nR\Delta T$$

$$C = \frac{\Delta Q}{n\Delta T} = \frac{\frac{f}{2}nR\Delta T + \frac{nR\Delta T}{1-a}}{n\Delta T} = \frac{f}{2}R + \frac{R}{1-a} = C_v + \frac{R}{1-a}$$

Q Find molar heat capacity of He gas, which follows the process $P^3V = const$

$P^3V = const$

Taking cube root

$PV^{1/3} = const$

$a = 1/3$

$$C = C_v + \frac{R}{1-a} = \frac{3}{2}R + \frac{R}{1-1/3}$$

$$\boxed{C = 3R}$$

Q Find molar heat capacity of O_2 gas, which follows the process $PT^2 = const$

$PT^2 = const$

$$P\left(\frac{PV}{nR}\right)^2 = const$$

$P^3V^2 = const$

$PV^{2/3} = const$

$a = 2/3$

$$C = C_v + \frac{R}{1-a}$$

$$C = \frac{5}{2}R + \frac{R}{1-2/3}$$

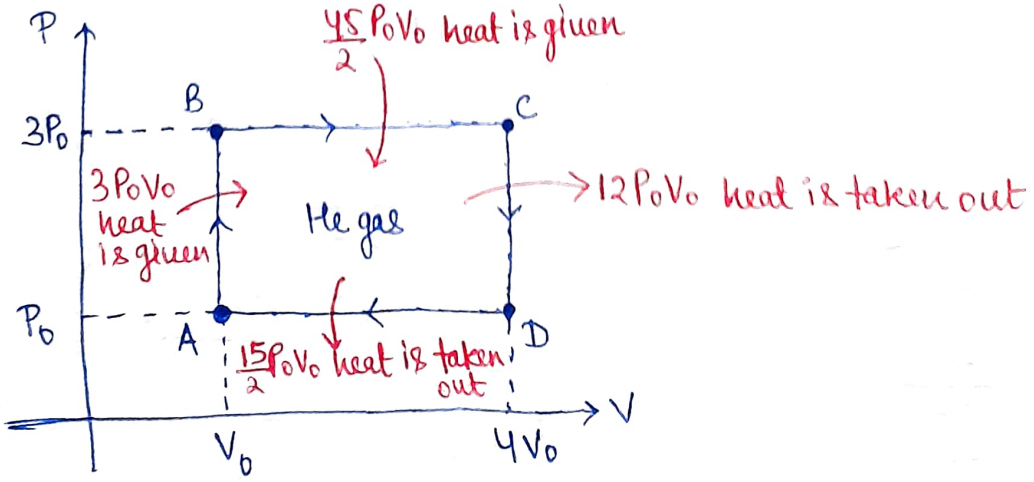
$$\boxed{C = \frac{11R}{2}}$$

Cyclic Process:

→ It is a combination of few processes

→ $P_f = P_i$, $V_f = V_i$, $T_f = T_i \Rightarrow \Delta U = 0$

Ex



	Process	W	$\Delta U = \frac{f}{2} nR\Delta T$	$\Delta Q = \Delta U + W$
$A \to B$	Isochoric	0	$\Delta U = \frac{3}{2}(P_f V_0 - P_i V_0)$ $\Delta U = \frac{3}{2} V_0 \Delta P = \frac{3}{2} V_0 (2P_0)$ $\Delta U = 3P_0V_0$	$3P_0V_0$
$B \to C$	Isobaric	$W = P\Delta V$ $W = 3P_0(4V_0 - V_0)$ $W = 9P_0V_0$	$\Delta U = \frac{3}{2}(3P_0V_f - 3P_0V_i)$ $\Delta U = \frac{9}{2} P_0 \Delta V = \frac{9}{2} P_0 (3V_0)$ $\Delta U = \frac{27}{2} P_0V_0$	$\frac{45}{2} P_0V_0$
$C \to D$	Isochoric	0	$\Delta U = \frac{3}{2}(V)(\Delta P) = \frac{3}{2} \times 4V_0 \times (-2P_0)$ $\Delta U = -12P_0V_0$	$-12P_0V_0$
$D \to A$	Isobaric	$W = P\Delta V$ $W = P_0(V_0 - 4V_0)$ $W = -3P_0V_0$	$\Delta U = \frac{3}{2} P\Delta V$ $\Delta U = \frac{3}{2} P_0(-3V_0) = \frac{-9}{2} P_0V_0$	$\frac{-15}{2} P_0V_0$
$A \to B \to C \to D \to A$	Cyclic	$6P_0V_0$ (Also area under P/V curve)	$\Delta U = 0$	$6P_0V_0$

Efficiency of cycle:

$$\boxed{\eta(\%) = \frac{W}{Q_{in}} \times 100} \quad (Q_{in} = \text{heat given})$$

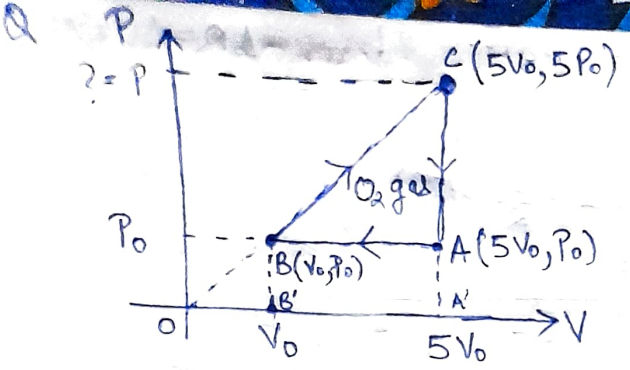
$$\eta = \frac{6P_0V_0}{(3 + \frac{45}{2})P_0V_0} \times 100 = \frac{12}{51} \times 100 = \frac{400}{17}\% = 23.52\%$$

Also, $\eta = \frac{W}{Q_{in}} \times 100 = \frac{\Delta Q}{Q_{in}} \times 100 = \frac{Q_{in} - Q_{out}}{Q_{in}} \times 100$

$$\boxed{\eta = \left(1 - \frac{Q_{out}}{Q_{in}}\right) \times 100}$$

$$\eta = \left(1 - \frac{(12 + \frac{15}{2})P_0V_0}{\frac{51}{2} P_0V_0}\right) \times 100 = \left(1 - \frac{39}{51}\right) \times 100$$

$$\eta = \frac{12}{51} \times 100 = \frac{400}{17}\% = 23.52\%$$



$\Delta OBB' \sim \Delta OCA'$

$\Rightarrow \frac{BB'}{CA'} = \frac{OB'}{OA'} \Rightarrow \frac{BB'}{OB'} = \frac{CA'}{OA'}$

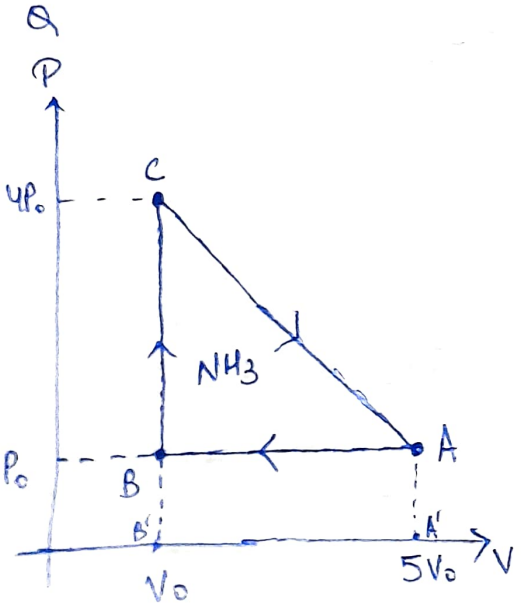
$= \frac{P_0}{V_0} = \frac{P}{5V_0} \Rightarrow \boxed{P = 5P_0}$

$\therefore$ P at C is $5P_0$

For process BC $\Rightarrow y = mx \equiv P = mV \Rightarrow PV^{-1} = \text{const} \Rightarrow a = -1$

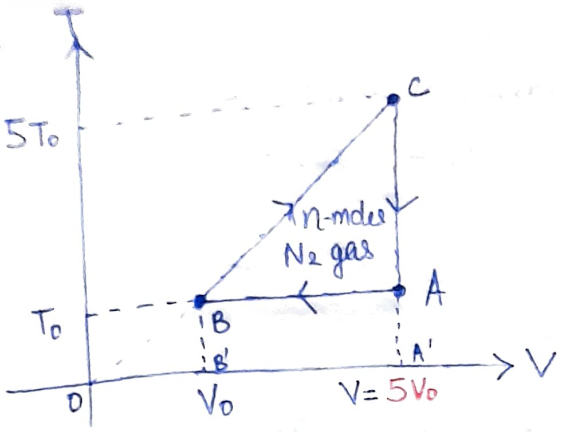
	Process	W	$\Delta U = \frac{f}{2} nR\Delta T$	$\Delta Q = \Delta U + W$
A→B	Isobaric	$W = P\Delta V$ $W = P_0(V_0 - 5V_0)$ $W = -4P_0V_0$	$\Delta U = \frac{5}{2} P_0 \Delta V = \frac{5}{2} P_0(-4V_0)$ $\Delta U = -10P_0V_0$	$-14P_0V_0$
B→C	Polytropic ($PV^{-1} = \text{const}$)	$W = \frac{(P_fV_f - P_iV_i)}{1-a}$ $W = \frac{25P_0V_0 - P_0V_0}{1-(-1)}$ $W = \frac{24P_0V_0}{2} = 12P_0V_0$	$\Delta U = \frac{5}{2}(P_fV_f - P_iV_i)$ $\Delta U = \frac{5}{2}(25P_0V_0 - P_0V_0)$ $\Delta U = 60P_0V_0$	$72P_0V_0$
C→A	Isochoric	0	$\Delta U = \frac{5}{2} V\Delta P$ $\Delta U = \frac{5}{2}(5V_0)(-4P_0)$ $\Delta U = -50P_0V_0$	$-50P_0V_0$
A→B→C→A	Cyclic	$8P_0V_0$	0	$8P_0V_0$

$\eta = \frac{W}{Q_{in}} \times 100 = \frac{8P_0V_0}{72P_0V_0} \times 100 = \frac{100}{9}\% = 11.1\%$



	Process	W	ΔU	ΔQ
A→B	Isobaric	$W = P\Delta V$ $W = P_0(-4V_0)$ $W = -4P_0V_0$	$\frac{6}{2} P_0(-4V_0)$ $\Delta U = -12P_0V_0$	$-16P_0V_0$
B→C	Isochoric	$W = 0$	$\frac{6}{2} V_0(4P_0 - P_0)$ $\Delta U = 9P_0V_0$	$9P_0V_0$
C→A		Area under CA = Area CA'A'BC $= \frac{1}{2}(4P_0 + P_0) \times 4V_0$ $= 10P_0V_0$	$\frac{6}{2}(5P_0V_0 - 4P_0V_0)$ $3P_0V_0$	$13P_0V_0$
A→B→C→A	Cyclic	$6P_0V_0$	0	$6P_0V_0$

Q



$\Delta OBB' \sim \Delta OCA'$

$$\frac{T_0}{V_0} = \frac{5T_0}{V}$$

$$\boxed{V = 5V_0}$$

For process BC ⇒

$PV = nRT \Rightarrow T = \frac{PV}{nR}$

Also $T \propto V \Rightarrow P = \text{const} \Rightarrow$ isobaric

	Process	W	ΔU	ΔQ
A → B	Isothermal	$nRT_0 \ln\left(\frac{V_0}{5V_0}\right)$ $= -nRT_0 \ln 5$	0	$-nRT_0 \ln 5$
B → C	Isobaric	$W = nR\Delta T$ $W = nR(4T_0)$	$\frac{5}{2} \times nR \times 4T_0$ $\Delta U = 10nRT_0$	$14nRT_0$
C → A	Isochoric	0	$\frac{5}{2} \times nR \times (-4T_0)$ $\Delta U = -10nRT_0$	$-10nRT_0$
A → B → C ← A	Cyclic	$nRT_0(4-\ln 5)$	0	$nRT_0(4-\ln 5)$

#Note: We cannot use area under curve to find work done here as this T V/s V graph not P V/s V graph.

$$\eta = \frac{W}{Q_{in}} \times 100 = \frac{nRT_0(4-\ln 5)}{14nRT_0} \times 100 = \frac{(4-\ln 5)}{14} \times 100\,\%$$

Adiabatic Process:

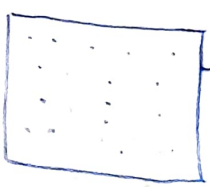
→ If gas expands or compress without exchanging heat with surrounding then process is adiabatic.

→ $dQ = 0$, $\Delta Q = 0$

→ $\Delta Q = \Delta U + W = 0 \Rightarrow \Delta U = -W$

→ If gas expands ⇒ $W = +ve$, $\Delta U = -ve \Rightarrow T\downarrow$

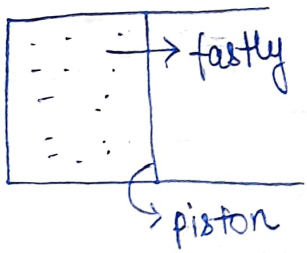
If gas compress ⇒ $W = -ve$, $\Delta U = +ve \Rightarrow T\uparrow$



→ non-conducting walls

⇓

adiabatic



⇒ adiabatic

Relation b/w P and V:

$dQ = 0 = dU + dW$

$0 = nC_v dT + PdV$

~~$0 = nC_v dT$~~

$0 = \frac{nRdT}{\gamma - 1} + PdV$

$PV = nRT$

Differentiate both sides

$PdV + VdP = nRdT$ — (1)

From (1)

$$\frac{PdV + VdP}{\gamma - 1} + PdV = 0 \Rightarrow \frac{PdV + VdP + \gamma PdV - PdV}{\gamma - 1} = 0$$

$= VdP + \gamma PdV = 0 \Rightarrow VdP = -\gamma PdV$

$= \int \frac{dP}{P} = -\gamma \int \frac{dV}{V} \Rightarrow \ln P = -\gamma \ln V + C$

$= \ln P + \gamma \ln V = C \Rightarrow \ln PV^\gamma = C$

$= PV^\gamma = e^C$

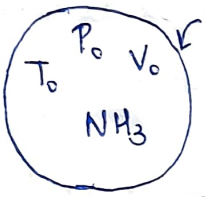
$\therefore$ **$PV^\gamma = \text{const}$** **$P_i V_i^\gamma = P_f V_f^\gamma$** **$\gamma = 1 + \frac{2}{f}$**

$\gamma_{mono} = 5/3$

$\gamma_{di} = 7/5$

$\gamma_{poly} = 4/3$

Q



This NH_3 gas expands from volume V_0 to $27V_0$. Find final pressure of NH_3 gas. Also find final temp.

(1) $P_i V_i^\gamma = P_f V_f^\gamma$

$= P_0 V_0^{4/3} = P(27V_0)^{4/3}$

$= P_0 V_0^{4/3} = P(27)^{4/3} V_0^{4/3}$

$= P = \frac{P_0}{(27)^{4/3}} = \frac{P_0}{3^4} = \frac{P_0}{81}$

(2) Moles conservation:

$$\frac{P_0 V_0}{T_0} = \frac{P_0 \times 27V_0}{81 \times T}$$

$$T = \frac{T_0}{3}$$

Now,

In adiabatic process,

$\Delta Q = \Delta U + W = 0$

$\Rightarrow W = -\Delta U$

$= -\frac{f}{2} nR\Delta T$

$= -nC_v \Delta T$

$= -\frac{nR\Delta T}{\gamma - 1}$

$$W = \frac{nR\Delta T}{1 - \gamma}$$

Relation b/w T and V

$PV^{\gamma} = \text{const}$

$\left(\frac{nRT}{V}\right)V^{\gamma} = \text{const}$

$\boxed{TV^{\gamma-1} = \text{const}}$

Relation b/w P and T

$PV^{\gamma} = \text{const}$

$P\left(\frac{nRT}{P}\right)^{\gamma} = \text{const}$

$\boxed{P^{1-\gamma}T^{\gamma} = \text{const}}$

Graph of P v/s V:

$PV^{\gamma} = \text{const}$

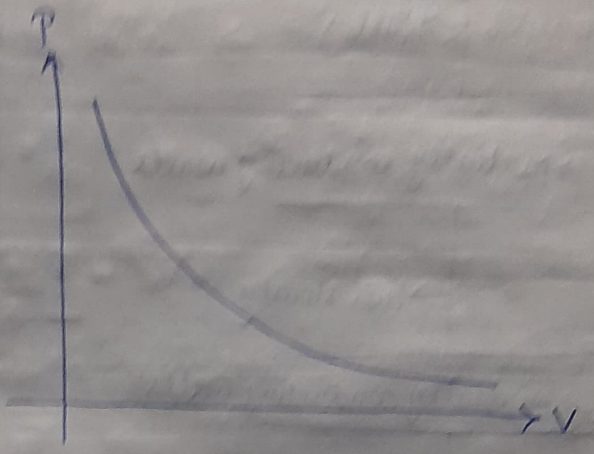
Differentiate both sides

$P\gamma V^{\gamma-1}dV + V^{\gamma}dP = 0$

$V^{\gamma-1}(P\gamma dV + VdP) = 0$

$\gamma PdV + VdP = 0$

$\boxed{\frac{dP}{dV} = -\frac{\gamma P}{V}}$



Comparison with isotherm:

Adiabatic: Slope $= \frac{dP}{dV} = -\frac{\gamma P}{V}$

Isothermal:

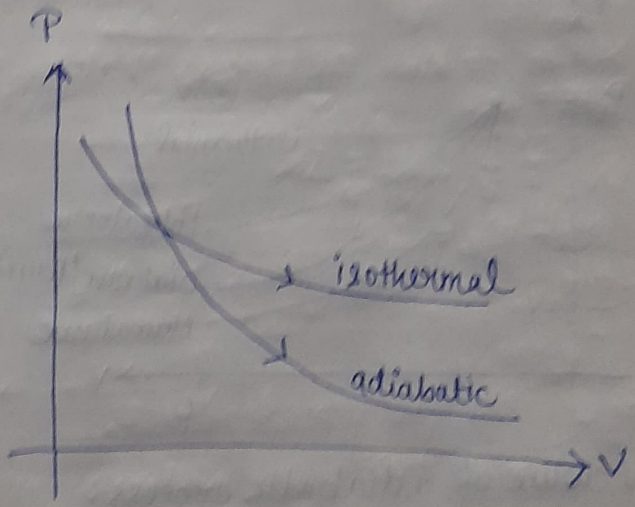
$PV = \text{const}$

$PdV + VdP = 0$

$PdV = -VdP$

$\boxed{\frac{dP}{dV} = -\frac{P}{V}}$

Slope $= \frac{dP}{dV} = -\frac{P}{V}$



Now we observe,

$\frac{(\text{Slope})_{\text{adiabatic}}}{(\text{Slope})_{\text{isothermal}}} = \gamma \quad (\gamma > 1)$

$\Rightarrow (\text{Slope})_{\text{Adiabatic}} > (\text{Slope})_{\text{isothermal}}$

$\therefore$ Adiabatic curve is more steeper than isotherm

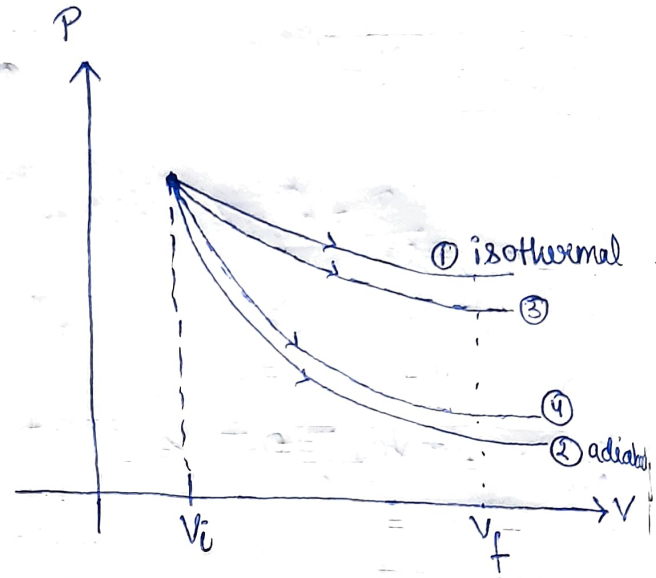
Note:

① Conducting walls → F_{ext} = slowly

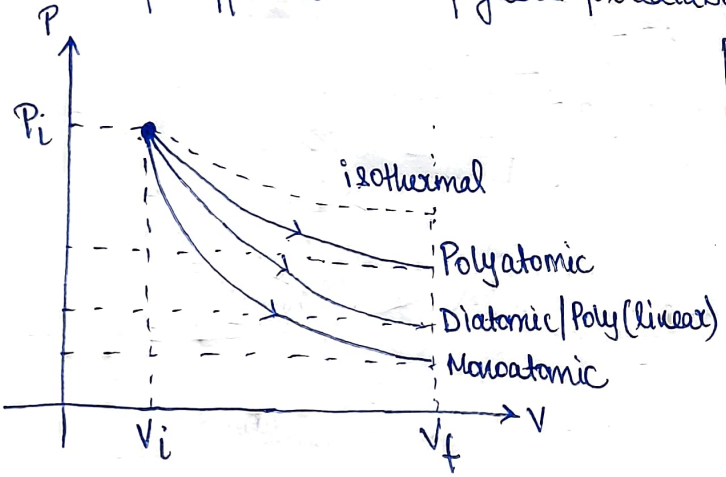
② non-conducting walls → Pull

③ partially conducting walls → Pull slowly

④ partially conducting walls → fastly



PV curve of different kind of gases for adiabatic process:



$$\frac{dP}{dV} = -\frac{\gamma P}{V}$$

$\gamma_{mono} > \gamma_{di} > \gamma_{poly}$

* If all gases are expanded to equal volume, their final pressures are.

$P_{poly} > P_{Dia} > P_{mono}$

$W_{poly} > W_{Dia} > W_{mono}$

TV curve of adiabatic process:

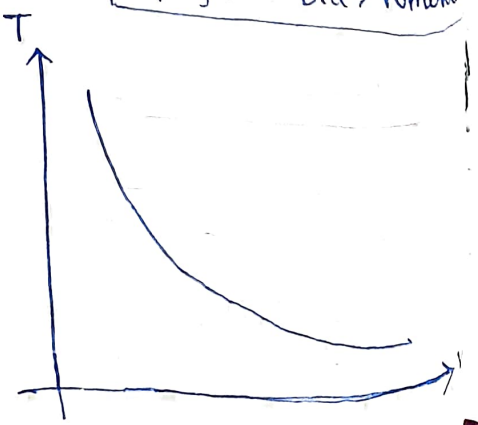
$$TV^{\gamma-1} = \text{const}$$

$$\Rightarrow (\gamma-1) T V^{\gamma-2} dV + V^{\gamma-1} dT = 0$$

$$\Rightarrow V^{\gamma-2} \{(\gamma-1) T dV + V dT\} = 0$$

$$\Rightarrow (\gamma-1) T dV + V dT = 0$$

$$\Rightarrow (\gamma-1) T dV = -V dT \Rightarrow \frac{dT}{dV} = -\frac{(\gamma-1)T}{V}$$



PT curve of adiabatic process:

$$P^{1-\gamma} T^{\gamma} = \text{const}$$

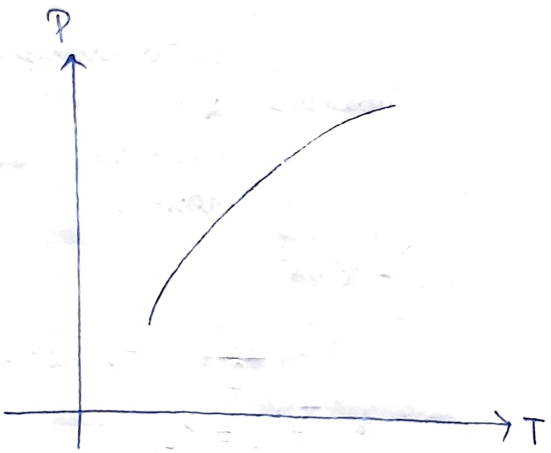
$$P^{1-\gamma}\gamma T^{\gamma-1} dT + T^{\gamma}(1-\gamma) P^{-\gamma} dP = 0$$

$$= (T^{\gamma-1} P^{-\gamma})\{P\gamma dT + T(1-\gamma)dP\} = 0$$

$$= \gamma P dT + (1-\gamma) T dP = 0$$

$$= \gamma P dT = (\gamma-1) T dP$$

$$\boxed{\frac{dP}{dT} = \frac{\gamma P}{(\gamma-1)T}}$$



Work done in adiabatic process:

$$PV^{\gamma} = \text{const} = K$$

$$P_i V_i^{\gamma} = P_f V_f^{\gamma} = K$$

$$W = \int P dV = \int \frac{K}{V^{\gamma}} dV$$

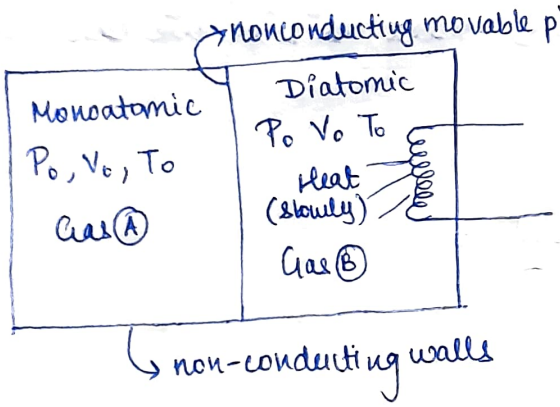
$$= \frac{K(V^{1-\gamma})_{V_i}^{V_f}}{1-\gamma} = \frac{K(V_f^{1-\gamma} - V_i^{1-\gamma})}{1-\gamma}$$

$$W = \frac{kV_f^{1-\gamma} - kV_i^{1-\gamma}}{1-\gamma} = \frac{P_f V_f^{\gamma} V_f^{1-\gamma} - P_i V_i^{\gamma} V_i^{1-\gamma}}{1-\gamma}$$

$$\boxed{W = \frac{P_f V_f - P_i V_i}{1-\gamma}}$$

$$\boxed{W = \frac{nR\Delta T}{1-\gamma}}$$

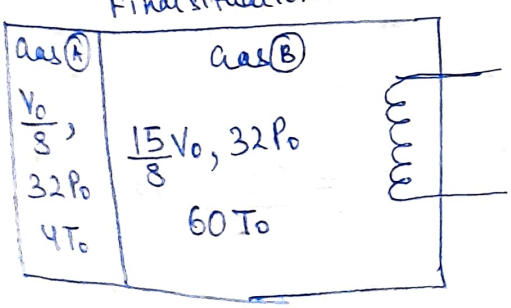
***Q



This arrangement is heated till volume of gas (A) becomes 12.5% of initial.

1. Find final volume, pressure and temp. of gas (A)
2. Find final volume, pressure and temp. of gas (B)
3. Find work done by gas (A) on piston.
4. Find work done by gas (B) on piston
5. Heat provided by heater.

Final situation.



(1) & (2): Final volume of gas A $\Rightarrow \frac{12.5}{100} \times V_0$

$$= \frac{V_0}{8}$$

Let final volume of gas B = V

$(V_i = V_f)$ $\quad 2V_0 = V + \frac{V_0}{8} \Rightarrow V = \frac{15V_0}{8}$

Now, since heater is delivering heat slowly, final pressure of both gases must be same as piston must be in equilibrium (Ex If heater raises temp of gas B, $P_B \uparrow$ so piston moves towards gas (A) due to which $V_A \downarrow$ which increases P_A and piston comes to equilibrium.)

Also, gas (A) can be said to go adiabatic compression (as no heat is exchanged due to non-conducting walls and piston and no heater for gas)

$$PV^\gamma = \text{const}$$

$$\Rightarrow P_0 V_0^{5/3} = P\left(\frac{V_0}{8}\right)^{5/3} \Rightarrow P_0 V_0^{5/3} = \frac{P V_0^{5/3}}{8^{5/3}} \Rightarrow P_0 = \frac{P}{32}$$

$$= P = 32P_0 = P_A = P_B$$

Also, $TV^{\gamma-1} = \text{const}$

$$T_0 V_0^{2/3} = T\left(\frac{V_0}{8}\right)^{2/3} \Rightarrow T_0 V_0^{2/3} = T\frac{V_0^{2/3}}{8^{2/3}} \Rightarrow T_0 = \frac{T}{4} \Rightarrow T = 4T_0 = T_A$$

For T_B we can use moles conservation:

$$\frac{P_0 V_0}{T_0} = \frac{32P_0 \cdot \frac{15V_0}{8}}{T_B} \Rightarrow \boxed{T_B = 60\,T_0}$$

(3) $$W_{adia} = \frac{P_f V_f - P_i V_i}{1-\gamma} = \frac{32P_0 \times \frac{V_0}{8} - P_0 V_0}{1-5/3} = \frac{-3P_0V_0}{2/3} = -\frac{9}{2}P_0V_0$$

(4) We do not know the process that gas (B) undergoes but we can use WET.

$$\text{WET} \Rightarrow W_{all} = \Delta K$$

$$W_A + W_B = \Delta K$$

$$-\frac{9}{2}P_0V_0 + W_B = 0 \qquad \therefore W_B = \frac{9}{2}P_0V_0$$

(5) By 1st law of thermodynamics,

$$\Delta Q = \Delta U + W$$

$$\Delta Q = \frac{5}{2}(60P_0V_0 - P_0V_0) + \frac{9}{2}P_0V_0$$

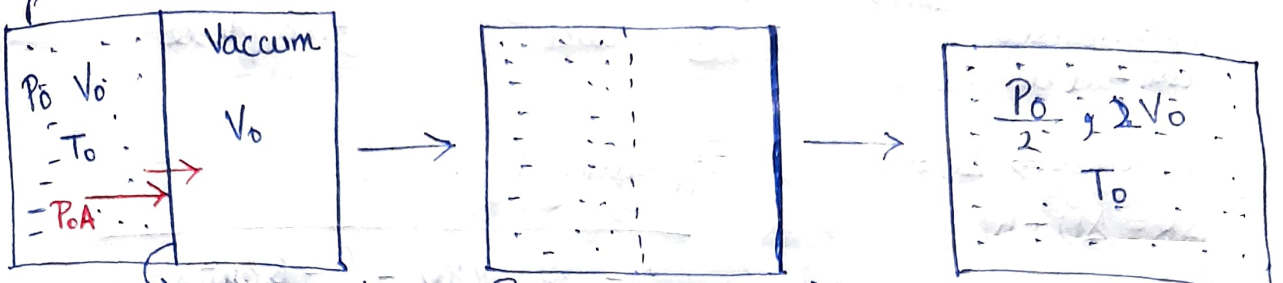
$$\Delta Q = \frac{59\times 5}{2}P_0V_0 + \frac{9}{2}P_0V_0 = \frac{(295+9)}{2}P_0V_0 = \frac{304}{2}P_0V_0$$

$$\Delta Q = 152P_0V_0$$

Free Expansion:

→ Gas is expanding against no force
→ Gas is expanding against zero pressure
→ Gas is expanding against vaccum.

① non-conducting walls



non-conducting, massless, movable piston

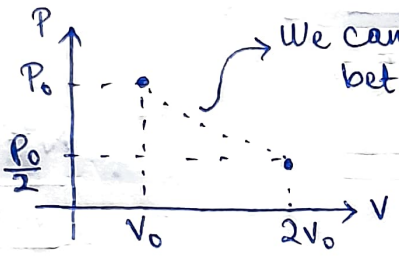
Piston moves rapidly to the other end due to impulse by P_0A. Gas now will occupy the remaining volume. At this stage we cannot define the P, V, T of gas

Final situation

→ $W_{gas} = 0$

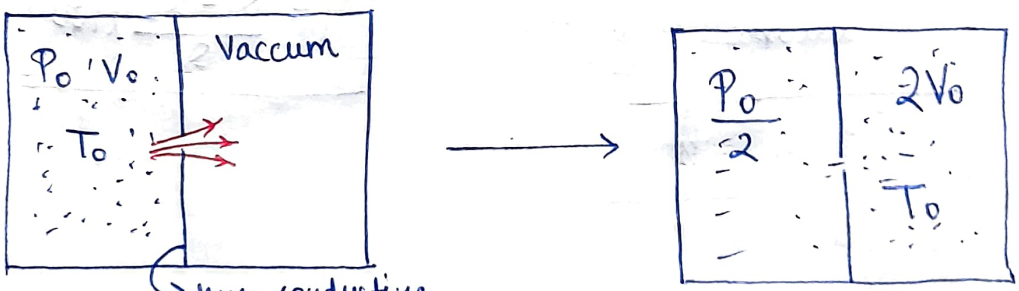
→ $\Delta Q = 0$

→ $\Delta Q = \Delta U + W \Rightarrow \Delta U = 0 \Rightarrow \Delta T = 0$



We cannot determine the graph in between as we have no idea about P, V in the intermediate stage.

②

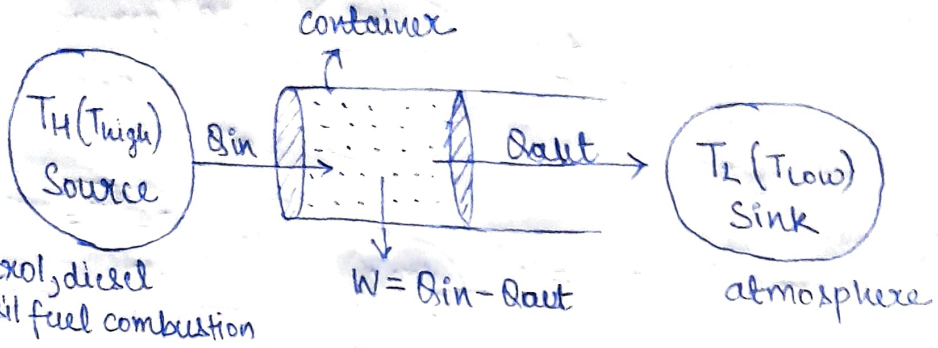


non-conducting, fixed piston

Heat Engine:

→ System which converts heat into mechanical energy

↳ heat into work done



Cyclic process $\Rightarrow \Delta U = 0$

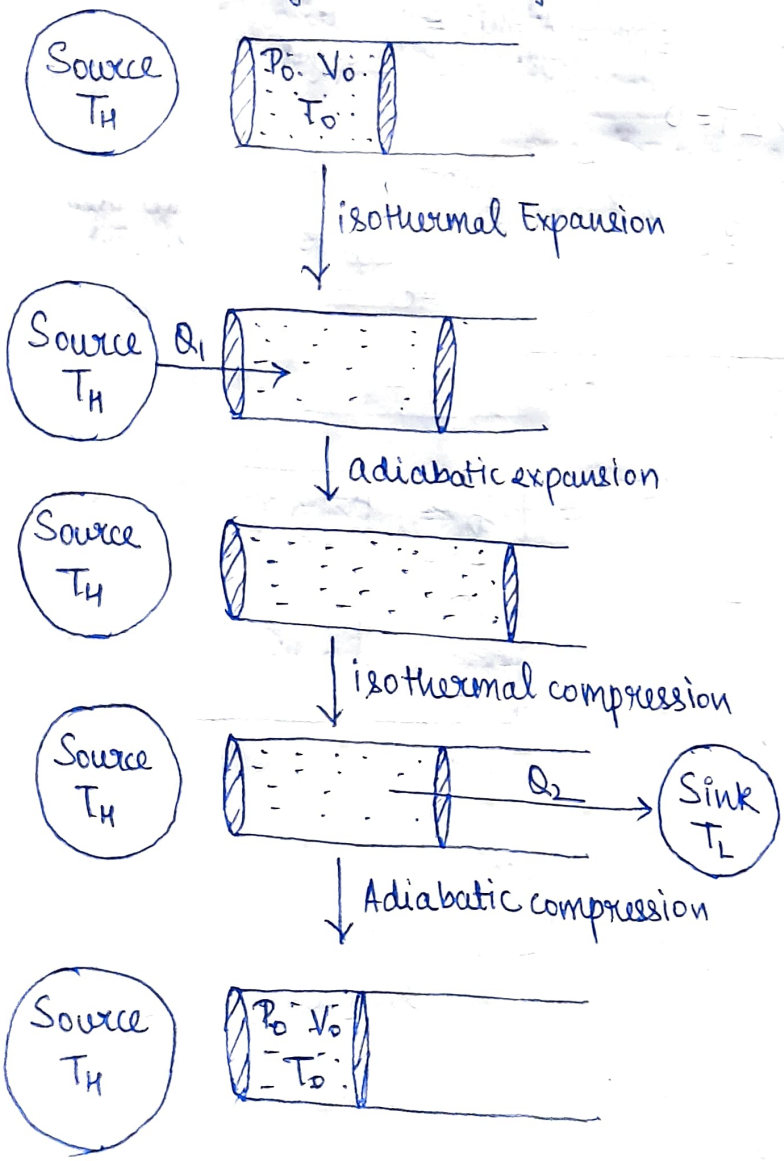
$\Delta Q = \Delta U + W$

$$W = Q_{in} - Q_{out}$$

Efficiency (η %):

$$\eta = \frac{W}{Q_{in}} \times 100 = \left(1 - \frac{Q_{out}}{Q_{in}}\right) \times 100$$

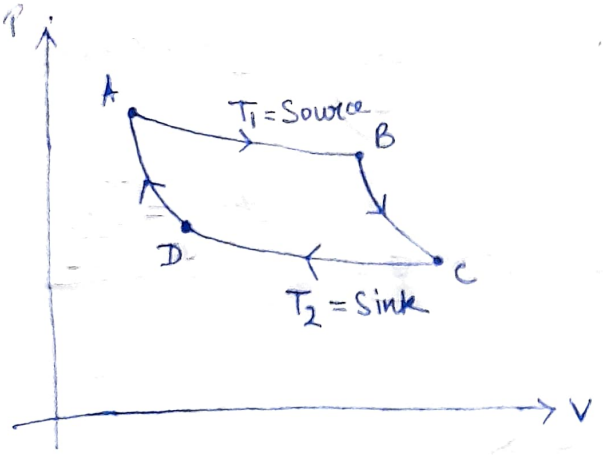
Carnot's Heat Engine (Carnot's cycle):



$W = Q_{in} - Q_{out}$

$$W = Q_1 - Q_2$$

$$\eta = \frac{Q_1 - Q_2}{Q_1} \times 100$$



A→B : isothermal expansion
B→C : Adiabatic expansion
C→D : isothermal compression
D→A : Adiabatic compression

Now,

① A→B : $\Delta U = 0$

$\Delta Q = W = nRT_1 \ln\left(\frac{V_B}{V_A}\right)$ - +ve.

$$\boxed{Q_{in} = nRT_1 \ln\left(\frac{V_B}{V_A}\right)}$$

② B→C : $\Delta Q = 0$

③ C→D : $\Delta U = 0$

$\Delta Q = W = nRT_2 \ln\left(\frac{V_D}{V_C}\right) = -ve$

$$\boxed{Q_{out} = nRT_2 \ln\left(\frac{V_C}{V_D}\right)}$$

④ D→A : $\Delta Q = 0$

Now,

$$\eta = \frac{W}{Q_{in}} \times 100 = \left(1 - \frac{Q_{out}}{Q_{in}}\right) \times 100 = \left\{1 - \frac{nRT_2 \ln\left(\frac{V_C}{V_D}\right)}{nRT_1 \ln\left(\frac{V_B}{V_A}\right)}\right\} \times 100 = \left\{1 - \frac{T_2 \ln\left(\frac{V_C}{V_D}\right)}{T_1 \ln\left(\frac{V_B}{V_A}\right)}\right\} \times 100$$

Now, we know B→C is adiabatic,

$\Rightarrow TV^{\gamma-1} = \text{const}$

$T_1 V_B^{\gamma-1} = T_2 V_C^{\gamma-1}$

$$\boxed{\frac{T_1}{T_2} = \left(\frac{V_C}{V_B}\right)^{\gamma-1}} \quad ①$$

Similarly for D→A:

$T_2 V_D^{\gamma-1} = T_1 V_A^{\gamma-1}$

$$\boxed{\frac{T_1}{T_2} = \left(\frac{V_D}{V_A}\right)^{\gamma-1}} \quad ②$$

From ① and ②

$\frac{V_C}{V_B} = \frac{V_D}{V_A}$

$\Rightarrow \frac{V_C}{V_D} = \frac{V_B}{V_A}$

$\Rightarrow \ln\left(\frac{V_C}{V_D}\right) = \ln\left(\frac{V_B}{V_A}\right)$

$\therefore \eta = \left\{1 - \frac{T_2 \ln\left(\frac{V_C}{V_D}\right)}{T_1 \ln\left(\frac{V_B}{V_A}\right)}\right\} \times 100 = \left(1 - \frac{T_2}{T_1}\right) \times 100 = \frac{T_1 - T_2}{T_1} \times 100$

$$\boxed{\eta = \frac{T_H - T_L}{T_H} \times 100 = \frac{T_{source} - T_{sink}}{T_{source}} \times 100}$$

Now, for $\eta = 100\%$. $\Rightarrow W = Q_{in}$

$\Rightarrow$ Heat is totally converting into work

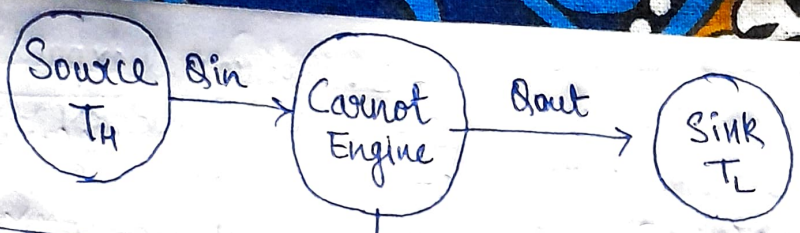
$\rightarrow T_2 = T_{sink} = 0$ Kelvin $\Rightarrow$ Not possible

$\therefore \eta = 100\%$ is not possible

Thermodynamics 2nd law :-

→ Efficiency of cycle can't be 100%.

→ Entire heat can't convert into mechanical energy or work.



$Q_{in} > Q_{out}$

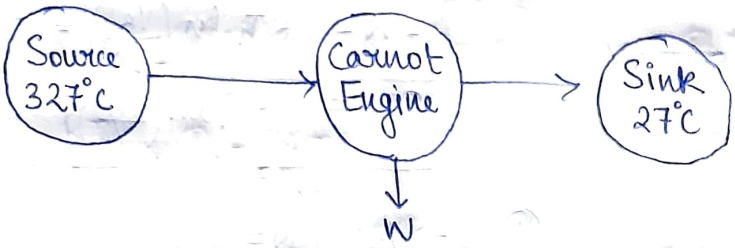
$$\eta = \frac{W}{Q_{in}} \times 100 = \frac{T_H - T_L}{T_H} \times 100$$



$W = Q_{in} - Q_{out}$

$$\frac{Q_{in}}{Q_{out}} = \frac{T_H}{T_L}$$

Q



Find

① Efficiency of cycle $\Rightarrow \eta = \frac{T_H - T_L}{T_H} \times 100 = \frac{600 - 300}{600} \times 100 = 50\%$.

② If heat extracted from source is 5000 cal, find work done and heat ejected to sink.

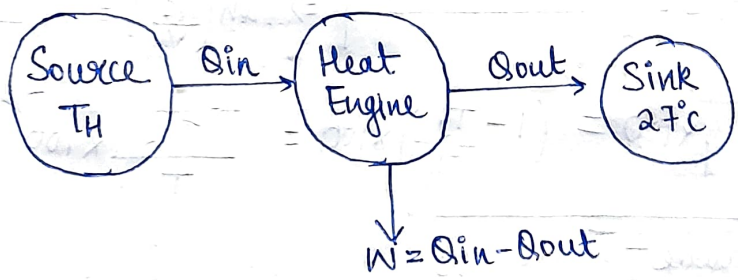
$$\eta = \frac{W}{Q_{in}} \times 100 \Rightarrow \frac{W}{5000} \times 100 = 50 \Rightarrow W = 2500 \text{ cal}$$

$Q_{sink} = 2500$ cal

③ If work done by this engine is to be 4kJ. Find heat extracted from source.

$$\eta = \frac{4kJ}{Q_{in}} \times 100 = 50 \Rightarrow Q_{in} = 8kJ$$

Q



Find temperature of source if efficiency is 40%.

$$\eta = \frac{T_H - 300}{T_H} \times 100 = 40 \Rightarrow 6T_H = 3000$$

$T_H = 500$ kelvin

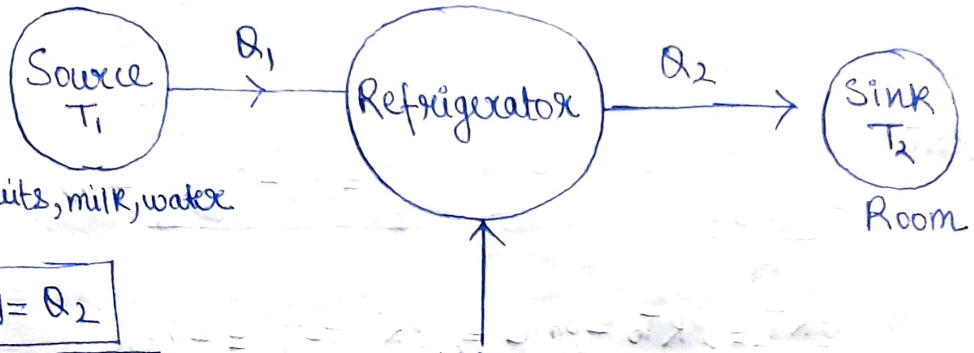
$$\eta_E = \frac{T_H - 300}{T_H} = \frac{40}{100}$$

$$= 10T_H - 3000 = 4T_H$$

Refrigerator / Reverse Heat Engine :-

→ Heat spontaneously flows from higher temperature to lower temperature. If we want to transfer heat from low temp. to high temp., work done by external agent is required.

In refrigerator, electrical energy supply source does work on it.



$$Q_1 + W = Q_2$$

$$W = Q_2 - Q_1$$

$$\frac{Q_1}{Q_2} = \frac{T_1}{T_2}$$

$$Q_1 < Q_2$$

$$T_1 < T_2$$

* If we open doors of refrigerator in a closed room, then room will heat instead of becoming cold as source will be room as well as sink will be room

$Q_{source} < Q_{sink}$

Coefficient of performance :

$$\beta = \frac{Q_1}{W} = \frac{Q_1}{Q_2 - Q_1} = \frac{T_1}{T_2 - T_1}$$

$$\therefore \beta = \frac{T_{inside}}{T_{out} - T_{inside}}$$

Q

Refrigerator
$-3^\circ C$

$T_{surr} = 27^\circ C$

Find

① Coefficient of performance :

$$\beta = \frac{T_{in}}{T_{out} - T_{in}} = \frac{270}{300 - 270} = \frac{270}{30} = 9$$

② Heat extracted is 1000 cal, then find w.d on refrigerator and heat dissipated by refrigerator to surrounding.

$$\beta = \frac{Q_{in}}{W} \Rightarrow 9 = \frac{1000}{W} \Rightarrow W = \frac{1000}{9} \text{ cal}$$

$$Q_2 = Q_1 + W$$

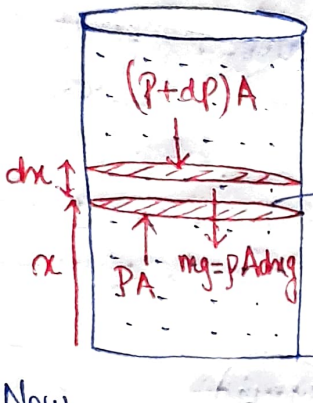
$$Q_2 = 1000 + \frac{1000}{9}$$

$$Q_2 = \frac{10000}{9} \text{ cal}$$

Alt $\frac{Q_1}{Q_2} = \frac{T_1}{T_2} \Rightarrow \frac{1000}{Q_2} = \frac{270}{309}$

$$Q_2 = \frac{10000}{9} \text{ cal}$$

Q



P is uniform

Find temp. of gas at height h from bottom.

For eqm:

$$(P+dP)A + \rho A g\,dx = PA$$

$$\int_{P_0}^{P} dP = -\rho g \int_0^h dx$$

$$P - P_0 = -\rho g h$$

$$\boxed{P = P_0 - \rho g h} \quad -(1)$$

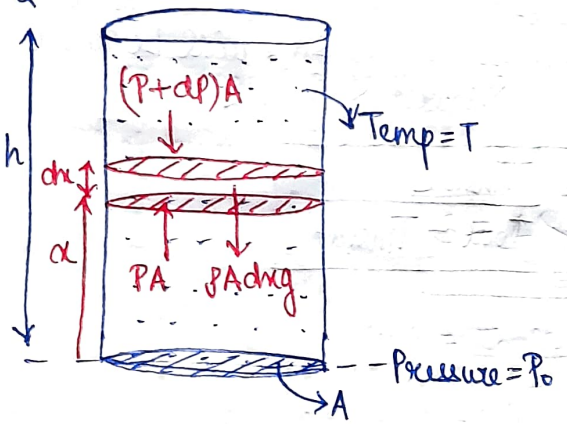
Now, $PM = \rho RT$

$$P = \frac{\rho RT}{M}, \quad P_0 = \frac{\rho R T_0}{M}$$

Put in (1) $\Rightarrow$ $\frac{\rho RT}{M} = \frac{\rho R T_0}{M} - \rho g h \Rightarrow \frac{\rho R}{M}(T - T_0) = -\rho g h$

$$T - T_0 = -\frac{Mgh}{R} \Rightarrow \boxed{T = T_0 - \frac{Mgh}{R}}$$

Q



Find mass of gas in the container.

$$(P+dP)A + \rho A\,dx\,g = PA$$

$$dP = -\rho g\,dx$$

$$= dP = -\frac{PM}{RT} g\,dx \qquad \left(PM = \rho RT,\ \rho = \frac{PM}{RT}\right)$$

$$= \int_{P_0}^{P} \frac{dP}{P} = -\frac{Mg}{RT}\int_0^x dx$$

$$= \ln P/P_0 = -\frac{Mgx}{RT}$$

$$\Rightarrow \boxed{P = P_0 e^{\frac{-Mgx}{RT}}}$$

Now, $\rho = \frac{PM}{RT} = \frac{P_0 M e^{\frac{-Mgx}{RT}}}{RT}$

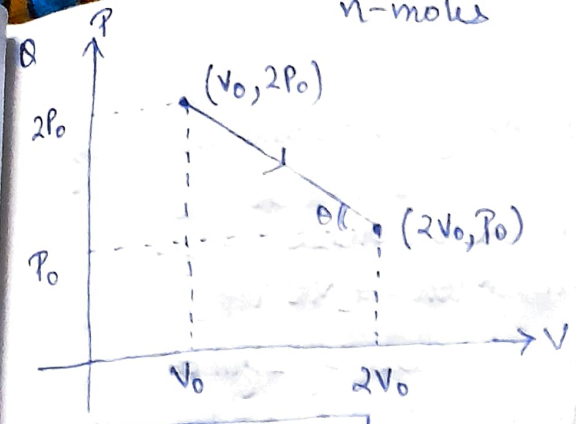
So, $dm = \rho\,dV$

$$dm = \frac{P_0 M \cdot e^{\frac{-Mgx}{RT}}}{RT} A\,dx$$

$$m = \int dm = \frac{P_0 MA}{RT}\int_0^h e^{\frac{-Mgx}{RT}}\,dx$$

$$m = \frac{P_0 MA}{RT}\left(\frac{\left(e^{\frac{-Mgx}{RT}}\right)_0^h}{\frac{-Mg}{RT}}\right) \Rightarrow \boxed{m = \frac{P_0 A}{g}\left(1 - e^{\frac{-Mgh}{RT}}\right)}$$

n-moles



Draw T v/s curve.
Find volume at which temp is maxm.
Also find maxm temp.

$y = mx + c$

$\equiv P = mV + c$

$\Rightarrow P = -\frac{P_0}{V_0}V + c \quad \{m = -\tan\theta\}$

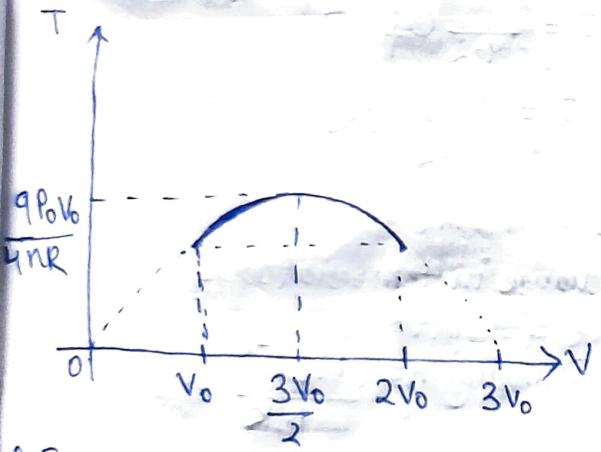
We know $(2V_0, P_0)$ satisfies the curve

$\Rightarrow P_0 = -\frac{P_0}{V_0}(2V_0) + c \Rightarrow \boxed{c = 3P_0}$

$\therefore \boxed{P = -\frac{P_0}{V_0}V + 3P_0}$

We know $PV = nRT \Rightarrow T = \frac{PV}{nR} \Rightarrow T = \frac{\left(-\frac{P_0}{V_0}V + 3P_0\right)V}{nR}$

$\therefore \boxed{T = -\frac{P_0V^2}{nRV_0} + \frac{3P_0}{nR}V}$



$\frac{dT}{dV} = -\frac{P_0 2V}{nRV_0} + \frac{3P_0}{nR} \quad (1)$

For maxima $\Rightarrow \frac{dT}{dV} = 0$

$\frac{3P_0}{nR} = \frac{P_0 2V}{nRV_0} \Rightarrow \boxed{V = \frac{3V_0}{2}}$

↳ volume where temp is max

Max temp:

$T = -\frac{P_0}{nRV_0}\left(\frac{9V_0^2}{4}\right) + \frac{3P_0}{nR} \times \frac{3V_0}{2} = \frac{9}{4}\frac{P_0V_0}{nR}$

8. For O_2 gas temp. is varying with volume as $T = T_0e^{KV}$. Find molar heat capacity of O_2 gas for this process.

$C = \frac{dQ}{ndT} = \frac{dU + dW}{ndT}$

$dU = \frac{f}{2}nRdT = \frac{5}{2}nRdT \quad -(1)$

$dW = PdV$

$= \left(\frac{nRT}{V}\right)dV$

$dW = \left(\frac{nRT_0e^{KV}}{V}\right)dV \quad -(2)$

Now,

$T = T_0e^{KV}$

$dT = T_0e^{KV} \cdot KdV$

So, $C = \frac{dU + dW}{ndT}$

$C = \frac{\frac{5}{2}nRdT}{ndT} + \frac{\left(\frac{nRT_0e^{KV}}{V}\right)dV}{nT_0e^{KV}KdV}$

$\boxed{C = \frac{5R}{2} + \frac{R}{KV}}$